

CSS BOX-SIZING

What Difference Does It Actually Make in the Output?

1. The Main Difference

The biggest difference between content-box and border-box is the final size of the element after padding and border are added.

content-box → declared width/height = CONTENT only

border-box → declared width/height = CONTENT + PADDING + BORDER

2. Same CSS — Different Output

Suppose we create two boxes with exactly the same width, padding, and border.

```
width: 200px;
padding: 20px;
border: 5px solid black;
```

Only the box-sizing value is different.

3. Output with content-box

```
.box {
  width: 200px;
  padding: 20px;
  border: 5px solid black;
  box-sizing: content-box;
}
```

For content-box, the 200px is only the content width.

Content	200px
Left + right padding	20 + 20 = 40px
Left + right border	5 + 5 = 10px
FINAL WIDTH	200 + 40 + 10 = 250px

OUTPUT: The box becomes 250px wide, even though you wrote width: 200px.

4. Output with border-box

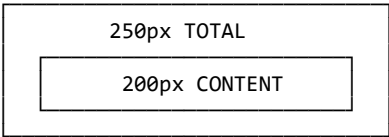
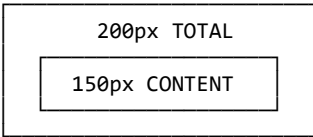
```
.box {
  width: 200px;
  padding: 20px;
  border: 5px solid black;
  box-sizing: border-box;
}
```

For border-box, the 200px is the complete outer width of the element.

Total width	200px
Left + right padding	40px
Left + right border	10px
Remaining content width	200 - 40 - 10 = 150px

OUTPUT: The box stays exactly 200px wide.

5. Visual Output Difference

content-box	border-box
	

6. Why This Matters in a Real Layout

The difference becomes very noticeable when several boxes are placed side-by-side. A box that becomes wider because of padding and border can push another box to the next line or make a layout overflow.

```
.left {  
  width: 50%;  
  padding: 20px;  
}  
  
.right {  
  width: 50%;  
  padding: 20px;  
}
```

If content-box is used, each 50% box becomes wider than 50% after padding is added. Together, they can exceed 100% of the available width.

Possible result: overflow, wrapping, or the second box moving below the first.

With border-box, the padding is included inside the declared 50% width, so the two boxes can fit much more predictably within the container.

This is one of the main practical reasons developers commonly use box-sizing: border-box.

7. Why We Used This in Your Lab 8 Code

```
* {  
  box-sizing: border-box;  
}
```

This rule applies border-box to every element. Therefore, when you specify a width or height for .page, .header, .content, .entry, or .sidebar, their padding and borders are included in those dimensions.

8. Width Calculation — Easy Formula

box-sizing	Total width
content-box	content + padding + border
border-box	declared width/height

9. In One Sentence

content-box makes padding and border increase the final size; **border-box** keeps the final outer size equal to the width/height you specified.

10. Quick Example to Remember

```
width: 200px;
padding: 20px;
border: 5px solid black;

content-box → 250px total
border-box  → 200px total
```

CSS • BOX-SIZING • OUTPUT DIFFERENCE